Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **BKD KUBO**
b. Team leader name: **Manuel III Ocaña**
c. Problem Statement: **BFSI: Intelligent Risk, Fraud & Financial Experiences**

Members: Manuel III Ocaña (Product Lead & Solution Architect) · Elson Mojica (Domain Lead, Finance & QA)

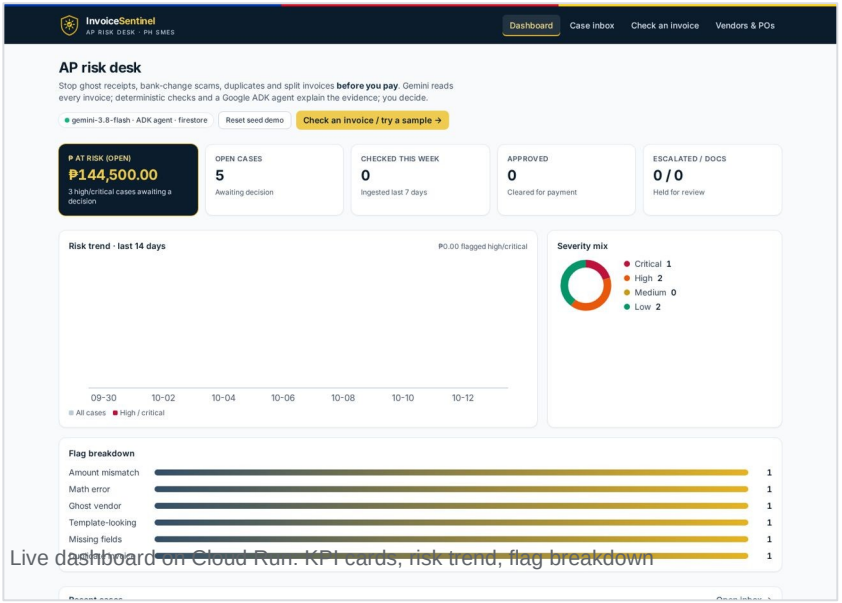
Solution: InvoiceSentinel, a pre-payment AP fraud & risk desk for PH SMEs

BKD KUBO

Brief about the idea

InvoiceSentinel checks every supplier invoice before it is paid and tells the AP analyst **PAY, HOLD or VERIFY**, with evidence.

- **Problem:** PH ghost receipts: the BIR alleges ₱25.5B in tax losses from 4 ghost firms (2019–21) and ₱1.41B against 23 buyer firms (2025). 68% of organizations saw an AP fraud attempt in 2024 (Basware).
- **Who:** SMEs with no fraud team, where one clerk reviews PDFs by eye.
- **How:** Gemini 3.8 Flash reads the invoice; deterministic code checks vendor master, PO, TIN, BIR RR 7-2024 and split invoices; a Google ADK agent explains in the user's language.
- **Human in the loop:** Approve / Escalate / Request docs, with an audit trail. Nothing is paid or sent automatically.



Live dashboard on Cloud Run: KPI cards, risk trend, flag breakdown

Opportunities

How is it different from existing ideas?

- OCR + rules tools need templates and can't judge "new bank, don't call us".
- Generic AI chat over a PDF has no vendor master, no PO, no audit trail.
- Opaque fraud scores can't be defended to auditors.
- InvoiceSentinel combines template-free Gemini reading with deterministic checks, evidence citations and a human decision.

How will it solve the problem?

- Flags the patterns that drain SMEs: ghost vendors, bank-account changes (BEC), duplicates, overbilling, PO mismatches, split invoices.
- Ranks the inbox by a deterministic 0–100 score, so high-risk invoices are reviewed first.
- Drafts a verification email to the contact on file, never the one on the invoice.
- BIR RR 7-2024 checklist keeps ghost receipts out of the books.

USP of the proposed solution

- **Code for facts, AI for judgment:** same input, same score (14 samples × 3 runs, zero variance).
- **Explainable agent:** Google ADK agent streams each tool step live and cites evidence IDs.
- **PH-first:** TIN, 12% VAT, BIR RR 7-2024, GCash/personal payees, Tagalog answers and emails.
- **Affordable:** ~₱0.31 Gemini cost per invoice.

List of features offered by the solution

Multimodal extraction
Gemini 3.8 Flash reads photos/PDFs into structured JSON (T=0, schema).

Vendor master & PO matching
Vendor, TIN, bank account, PO, quantities and prices; bank change = critical.

TIN validation
Format and branch-code checks, compared with the master record.

BIR RR 7-2024 checklist
Required invoice elements with section references; advisory only.

Split-invoice detection
Same vendor, close dates, combined total over the approval limit.

Deterministic risk score
Severity weights + floors; ranked inbox and ₱-at-risk KPIs.

Ask the case (Google ADK)
7 read-only tools; live streamed steps; answers end with PAY / HOLD / VERIFY.

Multilingual + verification email
Replies in English, Tagalog, etc.; email to the contact on file, never auto-sent.

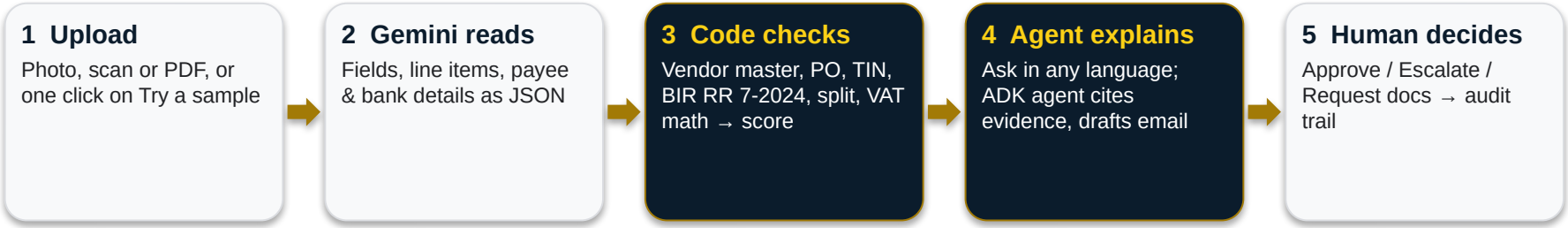
Try a sample
14 fictional PH invoices, one click, no files needed.

Decisions & audit trail
Approve / Escalate / Request docs with notes; timestamped log.

Vendors & POs page
Edit the master data the checks use; approval limit in Settings.

Accessible, responsive UI
AA contrast, keyboard & focus states, ARIA labels, mobile layouts.

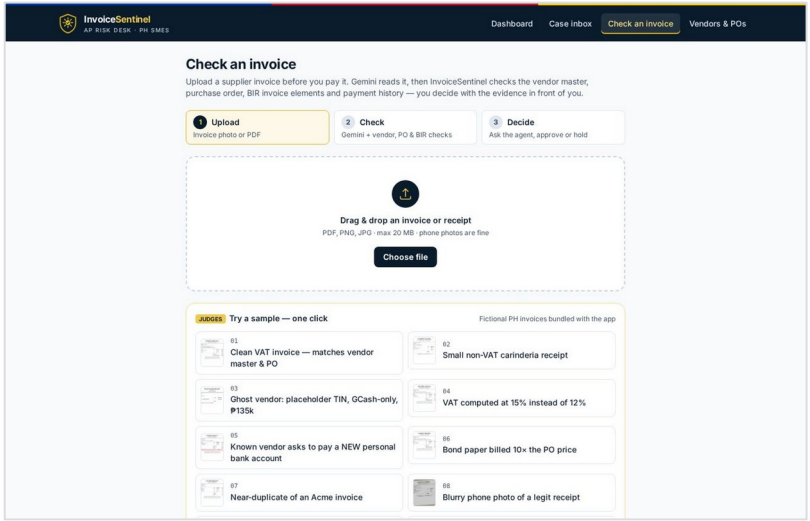
Process flow diagram or Use-case diagram



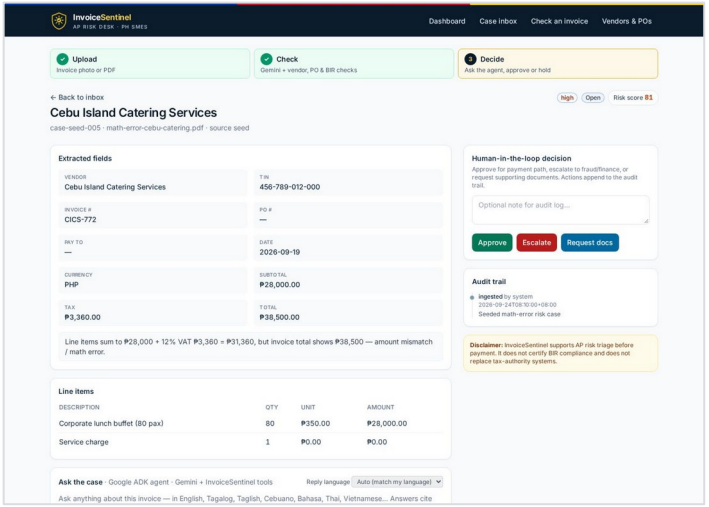
Use case: AP analyst at an SME receives invoice from known vendor “Northstar IT” (sample 09)

- Upload → Gemini extracts vendor, TIN, PO, totals and the bank account printed on the invoice.
- Vendor master check: everything matches except the bank account (invoice Metrobank vs master BPI) → UNUSUAL_PAYMENT_CHANGE, critical, score 90.
- Analyst asks in Tagalog “Ligtas ba itong bayaran?” → agent calls vendor, PO, history, TIN and BIR tools → answers HOLD with evidence.
- Analyst drafts a Tagalog verification email to the master contact, calls the supplier, and escalates; the decision is logged.

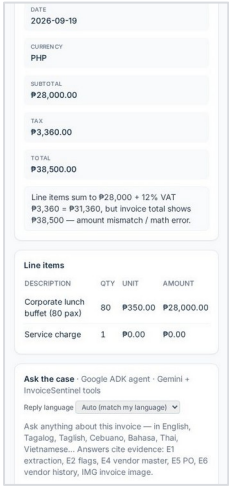
Wireframes/Mock diagrams of the proposed solution (optional)



Upload: guided stepper + Try a sample grid



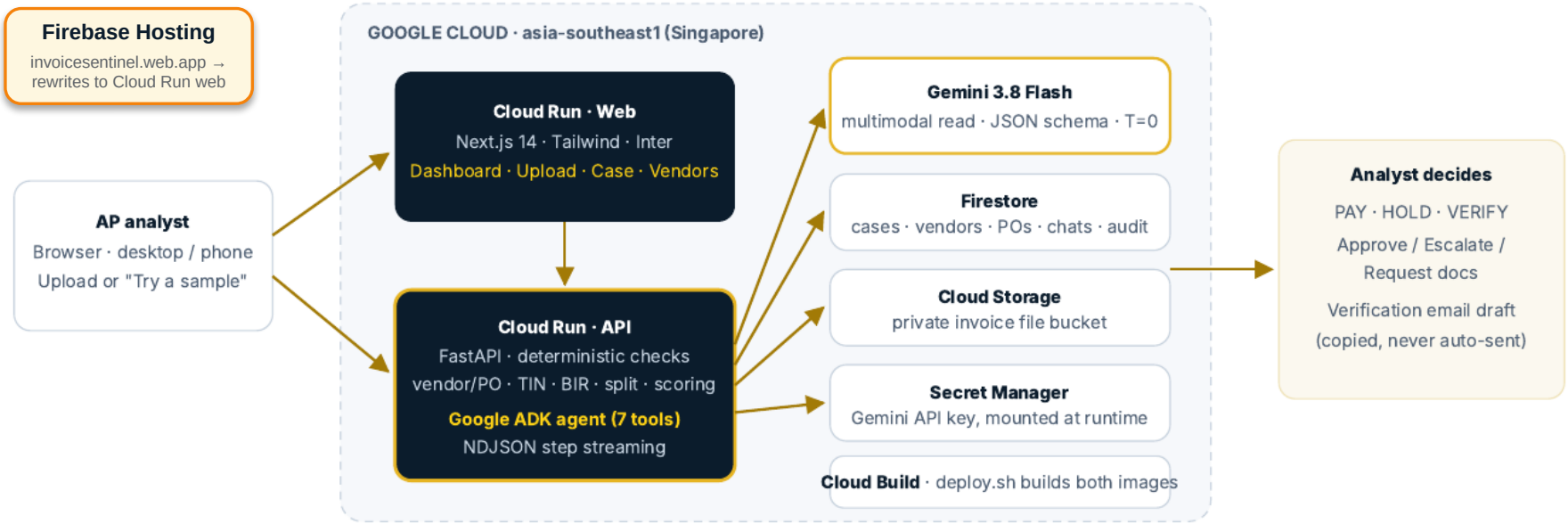
Case: fields, risk cards, matching, decision panel



Mobile layout

- Implemented in the live prototype (screens shown are real, not mockups).
- Flow: Dashboard → Check an invoice → Case → Decide; Vendors & POs page for master data.

Architecture diagram of the proposed solution



- Firebase Hosting serves invoicesentinel.web.app and rewrites all paths to the Cloud Run web service; the browser calls the API directly (no 60 s Hosting limit on agent streams).
- Stateless Next.js and FastAPI containers on Cloud Run (asia-southeast1); state in Firestore and a private Cloud Storage bucket.
- Gemini key in Secret Manager; one-command deploy.sh builds both images with Cloud Build.

Technologies to be used in the solution

Google AI

Gemini 3.8 Flash (google-genai SDK): multimodal read, JSON schema, T=0 · Google ADK 2.11: LlmAgent with 7 tools, streamed events

Google Cloud & Firebase

Firebase Hosting (invoicesentinel.web.app → Cloud Run) · Cloud Run (web + API, min-instances 1) · Firestore · Cloud Storage · Secret Manager · Cloud Build · Artifact Registry

Backend

Python 3.12 · FastAPI · Pydantic · deterministic rule engine (vendor/PO matching, TIN, BIR RR 7-2024, split detection, scoring) · NDJSON streaming

Frontend

Next.js 14 · React · TypeScript · Tailwind CSS · Inter · custom SVG charts · responsive, WCAG AA-minded UI

Quality

Stability harness (14 fictional samples × 3 runs) · sample generator · Playwright screenshots

Estimated implementation cost (optional)

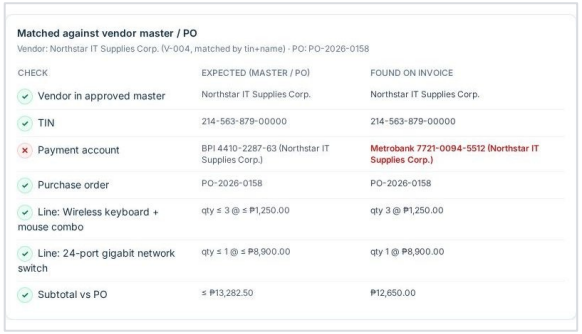
~₱0.31 per invoice

Gemini 3.8 Flash at 2026 promo pricing (~₱0.62 from Jan 2027). FX ₱62.77/US$.

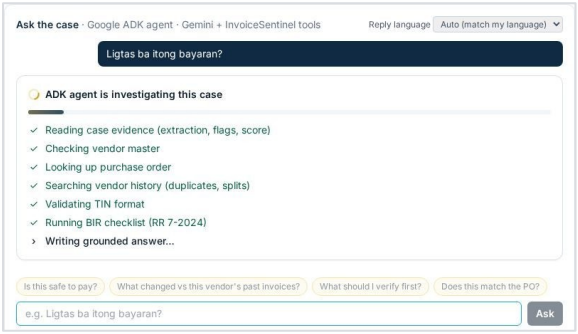
Invoices / month	Typical user	Model cost 2026	From 2027
200	Small SME	~₱62	~₱124
2,000	Mid-market finance team	~₱623	~₱1,246
50,000	Shared services / bank SME portfolio	~₱16,000	~₱31,000

- **Infrastructure:** Cloud Run scales to zero; at SME volumes usage fits the free tier. For judging, one warm instance per service (min-instances 1): ~US$23 (~₱1,450) per month; set to 0 afterwards.
- **Storage:** Firestore and Cloud Storage cost almost nothing at prototype scale.
- **Scale levers:** Gemini Batch API for overnight payment runs at half the token price; Flash by default.
- Source: Gemini Developer API pricing (updated Oct 9, 2026) and Cloud Run pricing. Per-invoice token use is an estimate.

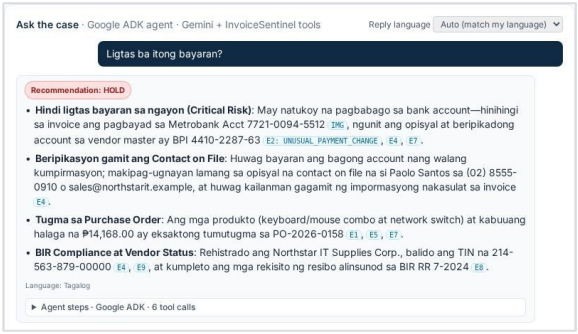
Snapshots of the prototype



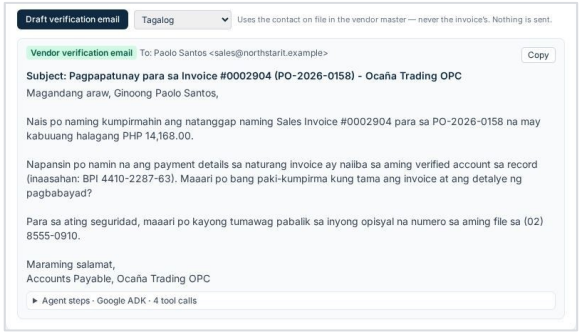
Vendor master: bank account mismatch (sample 09)



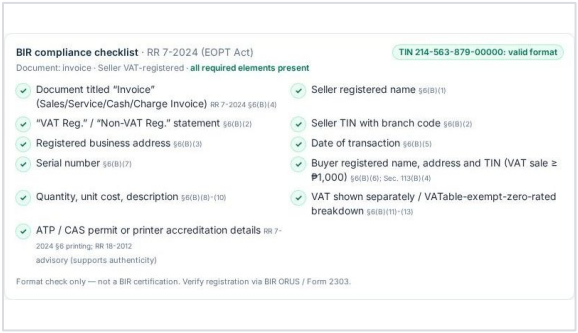
Google ADK agent steps streaming live



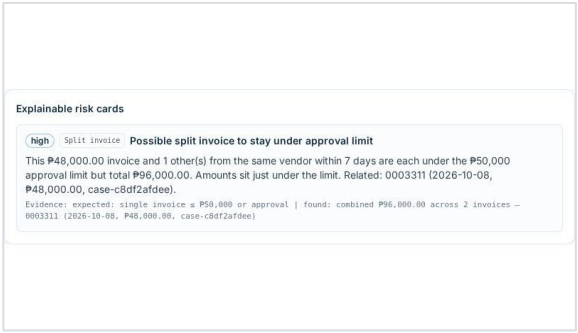
Tagalog answer: HOLD with cited evidence



Tagalog verification email to the contact on file

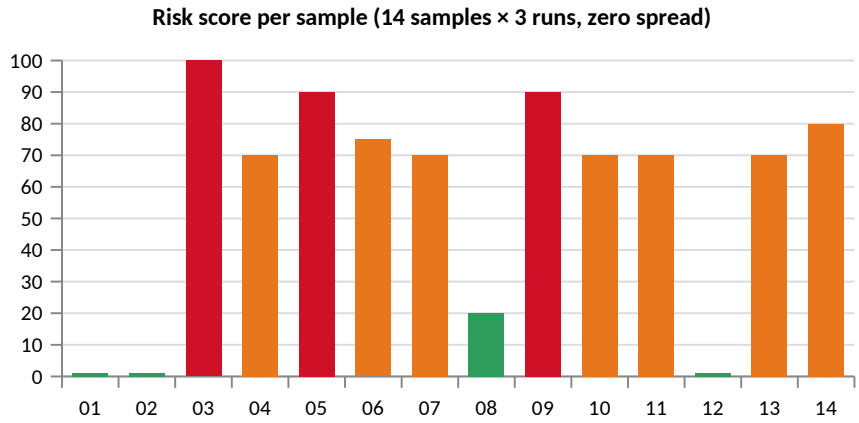


BIR RR 7-2024 checklist + TIN validation



Sample 13 flagged as a split of sample 12

Prototype Performance report/Benchmarking



Metric (live, Cloud Run)	Result
Score stability, 14 samples × 3 runs	0 spread, 14/14 in band
Upload → verdict (Gemini + checks)	~18–24 s
Agent: first streamed step	~5 s
Agent: full grounded answer	~18–37 s
Verification email draft	~20–25 s

Flag (expected samples)	Detected
GHOST_VENDOR (03, 14)	2 / 2
UNUSUAL_PAYMENT_CHANGE (05, 09)	2 / 2
TIN_INVALID (03, 14)	2 / 2
MATH_ERROR (04)	1 / 1
OVERBILLING (06)	1 / 1
DUPLICATE_INVOICE (07)	1 / 1
PO_MISMATCH (06, 10)	2 / 2
MISSING_PO (11)	1 / 1
SPLIT_INVOICE (13 after 12)	1 / 1
Clean invoices without fraud flags (01, 02, 08, 12)	4 / 4

Fictional PH test set, gemini-3.8-flash, Oct 12–13, 2026. Sample 08 (blurry photo) gets only a low-severity TIN-mismatch flag, not a fraud flag. Small test set: a labeled real-world evaluation set is on the roadmap.

Additional Details/Future Development (if any)

Q1 2027 · Pilot-ready

- Sign-in with analyst / approver roles (Identity Platform)
- Email-inbox intake of supplier bills
- CSV import of vendor master and POs
- Labeled PH evaluation set; precision/recall per flag

Q2 2027 · Integrations

- QuickBooks / Xero bill sync; hold payment on Escalate
- Batch review of payment runs (Gemini Batch API)
- Bank-account change history per vendor

Later · Ecosystem

- Pre-transfer check API for bank SME portals
- Claims-document review for insurers
- Tagalog / Cebuano UI
- BIR ATP / series lookups as signals

Guardrails stay fixed: a person decides every payment, every flag carries evidence, nothing is auto-sent, and the product never claims to certify BIR compliance. Paid-tier Gemini API: content is not used to train Google's models.

Provide links to your:

- **GitHub Public Repository**
  https://github.com/BKDKUBO/invoice-sentinel

- **Demo Video Link (3 Minutes)**
  [VIDEO LINK]

- **Final Product Link**
  https://invoicesentinel.web.app
  (Firebase Hosting → Cloud Run; direct: https://invoicesentinel-web-1034883854575.asia-southeast1.run.app)

Try it: Check an invoice → Try a sample → 09 (known vendor, new bank). Reset demo restores seed data. All data is fictional.

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Thank you

